Lecture 24: Optimal Linear Observer

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Hotelling observer
- Channelized linear observer

Optimal Linear Discriminant

We now consider discriminant functions $T(\cdot)$ that are optimal among all discriminant functions constrained to be linear in the data:

$$T(\mathbf{g}) = \mathbf{w}^T\mathbf{g}, \quad \text{where } \mathbf{g}, \mathbf{w} \in \mathbb{R}^M$$

- $\mathbf{w}$: Template (weight vector)

This form represents a linear weighted sum of the data elements, the random variables g_m. Linear discriminants are computationally efficient and easy to analyze. **The discrimenants maximize a certain measure of class separability**.

Hotelling Observer

The **Hotelling observer (HO)** is the *optimal linear observer* when both the first-order statistics ($\mathbf{E}_j[\mathbf{g}]$) and second-order statistics ($\mathbf{K}_j$) of the data are known. Among all linear discriminants, it maximizes the **SNR** (a measure of binary class separability).

$$\mathrm{SNR}_t = \frac{\mathrm{E}_2[t] - \mathrm{E}_1[t]}{\sqrt{\frac{1}{2}\sigma_1^2 + \frac{1}{2}\sigma_2^2}}$$

Hotelling Template

The linear discriminant (template) that maximizes SNR_t is given by:

$$\mathbf{w}_{Hot} = \mathbf{w}_{opt\ lin} = \mathbf{K}_\mathbf{g}^{-1}(\mathrm{E}_2[\mathbf{g}] - \mathrm{E}_1[\mathbf{g}]) = \mathbf{K}_\mathbf{g}^{-1}\Delta\mathrm{E}[\mathbf{g}]$$

The corresponding test statstic is:

$$t_{Hot} = T(\mathbf{g}) = \mathbf{w}_{Hot}{}^T\mathbf{g} = (\mathbf{K}_\mathbf{g}^{-1}\Delta\mathrm{E}[\mathbf{g}])^T\mathbf{g} = (\Delta\mathrm{E}[\mathbf{g}])^T\mathbf{K}_\mathbf{g}^{-1}\mathbf{g} \underset{D_1}{\overset{D_2}{\gtrless}} t_c$$

When the covariances under the two hypotheses differ, i.e., $\mathbf{K}_1 \neq \mathbf{K}_1$, the template that maximizes SNR becomes:

$$\mathbf{w}_{Hot} = \left[\frac{1}{2}(\mathbf{K}_1 + \mathbf{K}_2)\right]^{-1}\Delta\mathrm{E}[\mathbf{g}]$$

SNR

When the data have equal covariance $\mathbf{K}_\mathbf{g}$ under each hypothesis, the SNR for the HO becomes:

$$\mathrm{SNR}_{Hot}^2 = \mathrm{SNR}_{opt\ lin}^2 = (\Delta\mathrm{E}[\mathbf{g}])^T\mathbf{K}_\mathbf{g}^{-1}\Delta\mathrm{E}[\mathbf{g}]$$

When the **signal is known exactly** and the background is either known exactly or modeled as a Gaussian random process under both hypotheses:

$$H_1 : \mathbf{g} = \mathcal{H}\mathbf{f}_1 + \mathbf{n} \equiv \mathbf{b} + \mathbf{n}$$
$$H_2 : \mathbf{g} = \mathcal{H}\mathbf{f}_2 + \mathbf{n} \equiv \mathbf{s} + \mathbf{b} + \mathbf{n}$$
$$\mathrm{E}_1[\mathbf{g}] = \mathbf{b}, \quad \mathrm{E}_2[\mathbf{g}] = \mathbf{s} + \mathbf{b}, \quad \Delta\mathrm{E}[\mathbf{g}] = \mathrm{E}_2[\mathbf{g}] - \mathrm{E}_1[\mathbf{g}] = \mathbf{s}$$

The Hotelling SNR becomes:

$$\mathrm{SNR}_{Hot}^2 = \mathbf{s}^T\mathbf{K}_\mathbf{g}^{-1}\mathbf{s}$$

Relation between Hotelling and Ideal Observers

The Hotelling approach assumes a Gaussian model for the data, regardless of the true underlying distribution.

- If the data are *Gaussian with equal covariance across hypotheses*, the HO is equivalent to the ideal observer for the task, achieving the highest possible AUC.

- If the data are *non-Gaussian*, the ideal observer can be (usually is) nonlinear in the data, and HO may not acheive the highest possible AUC.

Required knowledge

- **Ideal observer**: Full probability density functions, i.e., likelihood functions $\mathrm{pdf}(\mathbf{g}|H_j)$, for the data
- **Hotelling observer**: Only the first-order statistics ($\mathrm{E}_j[\mathbf{g}]$) and second-order statistics ($\mathbf{K}_j$) for the data

Data Needed for Hotelling Observer Studies

For example,

- A flat-panel digital X-ray imager can have 1024 $\times$ 1024 detector elements.
- A SPECT system with a 128 $\times$ 128 detector collecting data over 64 projection angles results in the same number of elements.

In both cases, the data vector $\mathbf{g}$ has $M = 1,048,576$ elements, and the corresponding covariance matrix $\mathbf{K}_\mathbf{g} \in \mathbb{R}^{M\times M}$ contains over a trillion elements.

The number of values to be estimated to form the sample covariance matrix is almost always far greater than the number of samples available.

Given this dimensionality, **direct inversion** of the estimated covariance matrix is **not feasible**.

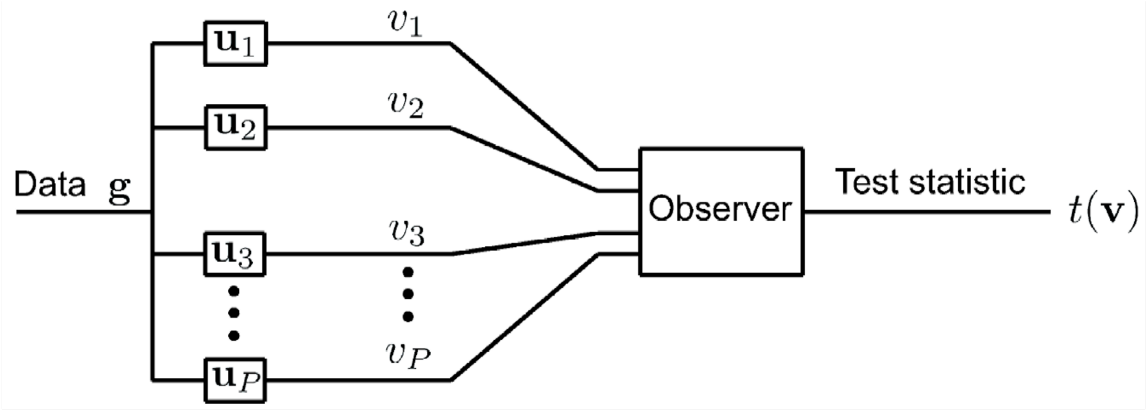
$$t_{Hot} = \mathbf{w}_{Hot}^T\mathbf{g} = (\mathrm{E}[\mathbf{g}])^T\mathbf{K}_\mathbf{g}^{-1}\mathbf{g}$$

Channelized Linear Observers

Any linear channel model can be represented using a marix-vector multiplication:

$$\mathbf{v} = \mathbf{U}^T\mathbf{g}, \quad \mathbf{U} = \begin{bmatrix} \vdots & \vdots & \vdots \\ \mathbf{u}_1 & \cdots & \mathbf{u}_P \\ \vdots & \vdots & \vdots \end{bmatrix}$$

- $\mathbf{U} \in \mathbb{R}^{M\times P}$: Matrix that consists of column vectors of the channel profiles $\mathbf{u}_p \in \mathbb{R}^M$
- $\mathbf{v} \in \mathbb{R}^P$: Vector of channel ouputs



Foundations of image science, Barrett and Myers

This transformation reduces the data dimensionality from M to P.

The channel profiles $\mathbf{u}_p$ can be *designed freely* to suit the purpose of the study.

Channels may be designed based on prior knowledge of the characteristics of the signal and background to minimize loss in detectability and approximate the inherent separability in the data.

The SNR for the channelized data $\mathbf{v}$ is given by:

$$\begin{aligned}\mathrm{SNR}_{\mathbf{v}}^2 &= (\mathrm{E}[\mathbf{v}])^T \mathbf{K}_{\mathbf{v}}^{-1} \Delta\mathrm{E}[\mathbf{v}] \\ &= (\mathrm{E}[\mathbf{U}^T\mathbf{g}])^T (\Delta\mathbf{v}\Delta\mathbf{v}^T)^{-1} \Delta\mathrm{E}[\mathbf{U}^T\mathbf{g}] \\ &= (\mathbf{U}^T\mathrm{E}[\mathbf{g}])^T [(\mathbf{U}^T\Delta\mathbf{g})(\mathbf{U}^T\Delta\mathbf{g})^T]^{-1} \Delta\mathrm{E}[\mathbf{U}^T\mathbf{g}] \\ &= (\mathrm{E}[\mathbf{g}])^T \mathbf{U} [\mathbf{U}^T\Delta\mathbf{g}\Delta\mathbf{g}^T\mathbf{U}]^{-1} \Delta\mathrm{E}[\mathbf{U}^T\mathbf{g}] \\ &= (\mathrm{E}[\mathbf{g}])^T \mathbf{U} [\mathbf{U}^T\mathbf{K}_{\mathbf{g}}\mathbf{U}]^{-1} \Delta\mathrm{E}[\mathbf{U}^T\mathbf{g}]\end{aligned}$$

Data Needed for Channelized Observer Studies

To evaluate the performance of a channelized observer, only the first- and second-order statistics of the data as seen through the channels, i.e., after projection onto the channel space, are required.

If the number of channels is P, then the covariance matrix $\mathbf{K}_{\mathbf{v}} \in \mathbb{R}^{P\times P}$, where $P \ll M$. As a result, the matrix to be inverted is of **manageable size**.

There is also **flexibility in choosing the channel profile**. Prior knowledge of the signal and background characteristics can guide the design of efficient channels, achieving $\mathrm{SNR}_{\mathbf{v}}^2 = \mathrm{SNR}_{\mathbf{g}}^2$.

Channel Example of a Detection Task

Consider a detection task in which:

- The signal is approximately **radially symmetric**, **centrally peaked** and **smooth**
- It is located at a **known position** on a **stationary background** with a correlation that has no preferred orientation.

In such settings, **Laguerre-Gauss (LG) functions** are suitable as channel profiles $\{u_p\}$ defined as:

$$u_p(r|a_u) = \frac{\sqrt{2}}{a_u}\exp\left(\frac{-\pi r^2}{a_u^2}\right)L_p\left(\frac{2\pi r^2}{a_u^2}\right)$$

- a_u: Scaling parameters, chosen based on prior knowledge of the signal and background
- $\exp\left(\frac{-\pi r^2}{a_u^2}\right)$: Gaussian factor
- $L_p\left(\frac{2\pi r^2}{a_u^2}\right)$: Laguerre polynomials

$$L_p(x) = \sum_{k=0}^{p}(-1)^k\binom{p}{k}\frac{x^k}{k!}$$

Orthogonality Relation

In 1D:

$$\int_0^\infty dx\, L_p(x)L_{p'}(x) = \delta_{pp'}$$

To a 2D (radial symmetry, with $x = \frac{2\pi r^2}{a_u^2}$):

$$\int_0^{2\pi} d\theta \int_0^\infty dr\, \frac{4\pi r}{a_u^2}\exp\left(-\frac{2\pi r^2}{a_u^2}\right)L_p\left(\frac{2\pi r^2}{a_u^2}\right)L_{p'}\left(\frac{2\pi r^2}{a_u^2}\right) = \delta_{pp'}$$

Thus, the LG functions form an orthogonal basis for radially symmetric functions in 2D.

Function Expansion with LG Basis

Any radially symmetric function $f(r)$ can be expanded in the LG basis:

$$f(r) = \sum_{p=0}^{\infty}\alpha_p u_p(r|a_u) = \sum_{p=0}^{\infty}\alpha_p\frac{\sqrt{2}}{a_u}\exp\left(\frac{-\pi r^2}{a_u^2}\right)L_p\left(\frac{2\pi r^2}{a_u^2}\right)$$

$$= \frac{\sqrt{2}}{a_u}\exp\left(\frac{-\pi r^2}{a_u^2}\right)\sum_{p=0}^{\infty}\alpha_p L_p\left(\frac{2\pi r^2}{a_u^2}\right)$$

where:

$$\alpha_p = \langle u_p(r), f(r) \rangle = \int d^2r \, u_p(r) \, f(r)$$

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from math import comb, factorial

def lg_channel(p, r, a_u):
    # Laguerre-Gauss radial profile (non-normalized for display)
    x = (2*np.pi*r**2)/(a_u**2)
    gaussian_factor = np.exp((-np.pi*r**2)/(a_u**2))
    L = np.zeros_like(x)
    for k in range(p + 1):
        L += (-1)**k * comb(p, k) * (x**k)/factorial(k)

    return (np.sqrt(2)/a_u)*gaussian_factor*L

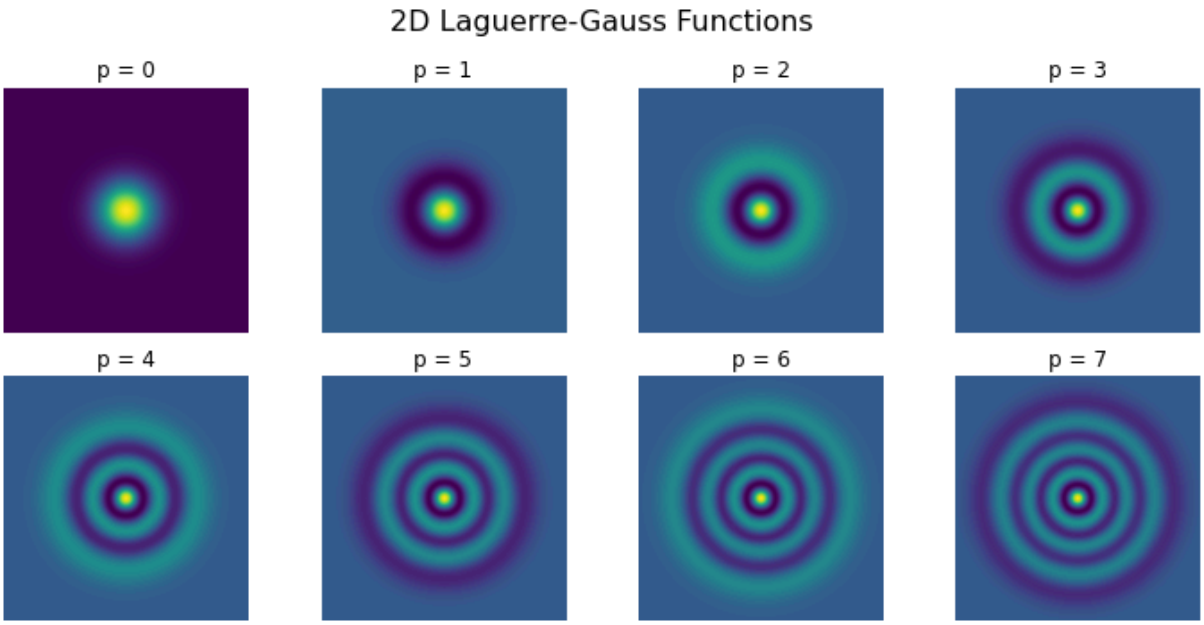
# Parameters
a_u = 0.4          # Scaling parameter
num_channels = 6   # Number of LG channels to visualize
grid_size = 128    # image resolution

# Create a polar grid
x = np.linspace(-1, 1, grid_size)
y = np.linspace(-1, 1, grid_size)
xx, yy = np.meshgrid(x, y)
rr = np.sqrt(xx**2 + yy**2)
```

In [2]:
```
# Plot
fig, axes = plt.subplots(2, 4, figsize=(10, 5))
fig.suptitle('2D Laguerre-Gauss Functions', fontsize=16)

for p, ax in enumerate(axes.flat):
    img = lg_channel(p, rr, a_u)
    ax.imshow(img, extent=(-1, 1, -1, 1), origin='lower')
    ax.set_title(f'p = {p}')
    ax.axis('off')

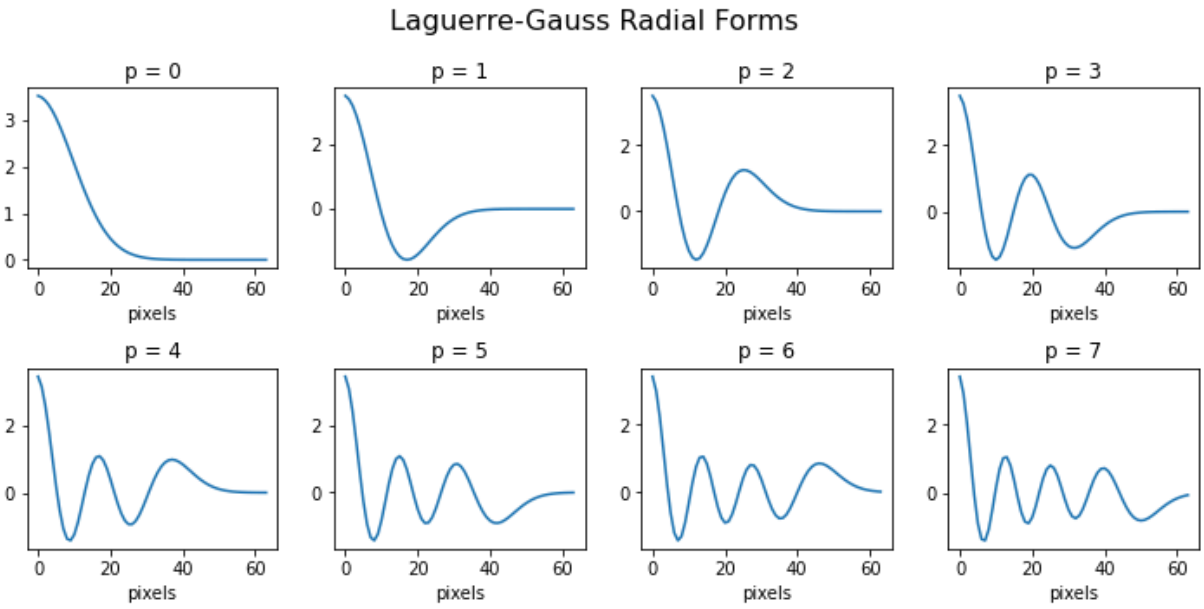
plt.tight_layout()
plt.show()
```



In [3]:
```
# Plot
fig, axes = plt.subplots(2, 4, figsize=(10, 5))
fig.suptitle('Laguerre-Gauss Radial Forms', fontsize=16)

for p, ax in enumerate(axes.flat):
    img = lg_channel(p, rr, a_u)
    ax.plot(img[grid_size//2:,grid_size//2])
    ax.set_title(f'p = {p}')
    ax.set_xlabel('pixels')

plt.tight_layout()
plt.show()
```



📖 *Gallas and Barrett (2003) investigated a signal-known-exactly (SKE) detection task on a lumpy background* [DOI: 10.1364/JOSAA.20.001725].

- A range of values for P can be investigated to determine the point at which detectability is maximized.
- The finding show that the performance of the channelized linear observer closely matches that of the ideal linear observer, using only 5 to 30 channels, depending on the complexity of the background statistics.

❓ How does the number of samples affect the reliability of the estimated covariance matrix?

❓ Why is this important for the performance of model observers such as the Hotelling observer?

In [22]:
```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

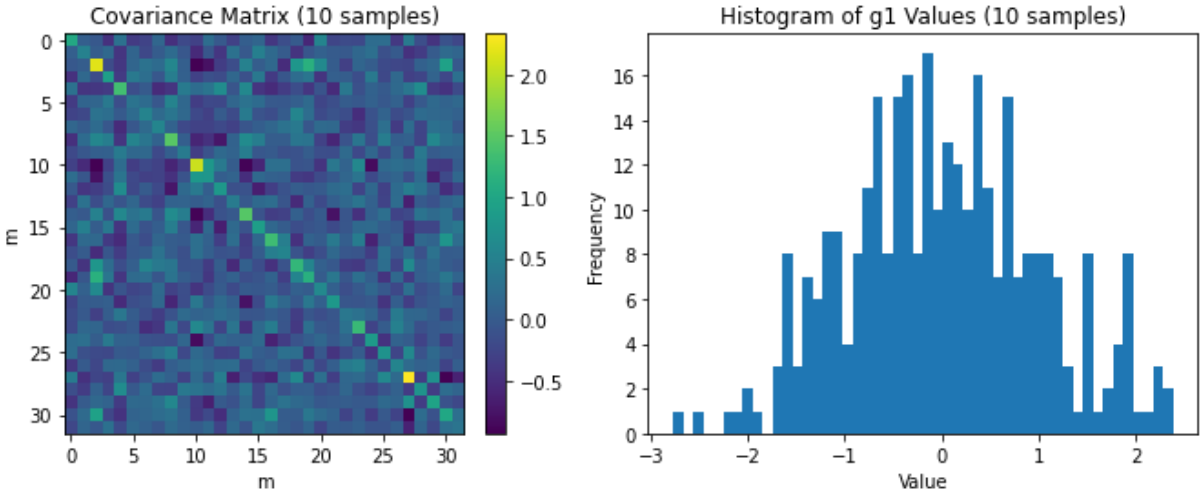
mean = 0
std = 1

# Case 1: 10 samples
g1 = np.random.normal(mean, std, (32, 10)) # M = 32, 10 samples
K1 = np.cov(g1)

# Case 2: 10,000 samples
g2 = np.random.normal(mean, std, (32, 10000)) # M = 32, 10,000 samples
K2 = np.cov(g2)
```

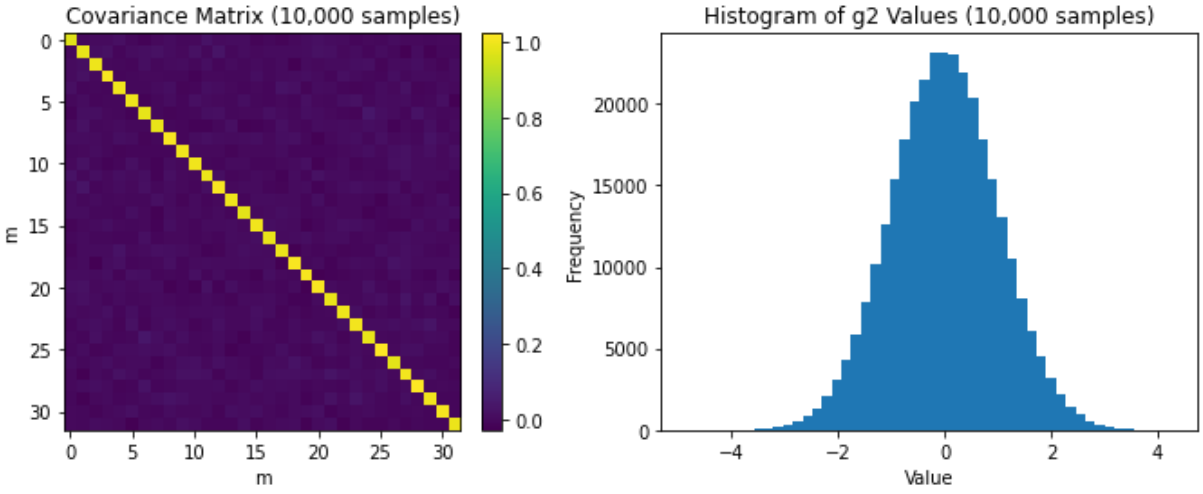
In [31]:
```
fig, axs = plt.subplots(1, 2, figsize=(10, 4))
im1 = axs[0].imshow(K1, cmap='viridis')
axs[0].set_title('Covariance Matrix (10 samples)')
axs[0].set_xlabel('m')
axs[0].set_ylabel('m')
fig.colorbar(im1, ax=axs[0], fraction=0.046, pad=0.04)

axs[1].hist(g1.flatten(), bins=50)
axs[1].set_title('Histogram of g1 Values (10 samples)')
axs[1].set_xlabel('Value')
axs[1].set_ylabel('Frequency')
plt.tight_layout()
plt.show()
```



In [30]:
```
fig, axs = plt.subplots(1, 2, figsize=(10, 4))
im2 = axs[0].imshow(K2, cmap='viridis')
axs[0].set_title('Covariance Matrix (10,000 samples)')
axs[0].set_xlabel('m')
axs[0].set_ylabel('m')
fig.colorbar(im2, ax=axs[0], fraction=0.046, pad=0.04)

axs[1].hist(g2.flatten(), bins=50)
axs[1].set_title('Histogram of g2 Values (10,000 samples)')
axs[1].set_xlabel('Value')
axs[1].set_ylabel('Frequency')
plt.tight_layout()
plt.show()
```



Errors in Estimates of SNR for Linear Observers

When the SNR is estimated from a finite number of samples, it may differ from the true SNR that would be obtained with full knowledge of the ensemble statistics. These estimates can exhibit both bias and variance.

In the case of the CHO, the SNR estimate is often highly biased unless the number of training samples is much greater than the number of channels.

Training and Testing

In practice, the observer template $\mathbf{w}$ is estimated from sample data and then applied to a separate set of data to evaluate performance.

- A **training set** is used to estimate the observer template, choose the number of channels, assign weights, and tune parameters of the channel profiles.
- A **testing set** is used to assess observer performance.

This approach is known as the *holdout* method, and the dataset split does not need to be balanced in size.

Training corresponds to defining the channel matrix $\mathbf{U}$ (in the case of CHO) and estimating the observer template $\mathbf{w}$ from the training data.